A MINI PROJECT REPORT ON

# GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform

*submitted in partial fulfillment of the requirements*
*for the award of the Degree of*

**BACHELOR OF TECHNOLOGY**
**in**
**COMPUTER ENGINEERING**

*by*


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**Class: T.E. Computer Engineering | Divisions: C & D**

*under the guidance and mentorship of*

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**Academic Year: 2026-2027**

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# CERTIFICATE OF APPROVAL

This is to certify that the Mini Project entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" is a bonafide work carried out by:

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in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Engineering (Third Year Computer Engineering, Divisions C & D) from the University of Mumbai during the Academic Year 2026-2027.

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# PROJECT APPROVAL CERTIFICATE

This is to certify that the Mini Project entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" submitted by Vedant Patil (2024300170), Ashmit Patil (2024300168), and Mayuri Tarde (2024300258) has been duly examined and approved for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Engineering from the University of Mumbai.

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| **Internal Examiner / Guide** | **External Examiner** |
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_____
**Seal of the Institute**

# CANDIDATES' DECLARATION

We hereby declare that the work presented in this Mini Project Report entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" is an authentic record of our own research, system implementation, and experimental benchmarking carried out as requirements for the completion of the Mini Project (Semester VI) in Bachelor of Technology in Computer Engineering at Sardar Patel Institute of Technology, Mumbai, under the guidance and mentorship of Dr. Rupali Sawant.

**We solemnly confirm that:**

- **1.** The matter embodied in this report has not been submitted by us for the award of any other degree or diploma to any other University or Institute.
- **2.** All literary contributions, algorithms, and architectural frameworks referenced from other authors have been explicitly cited and acknowledged in IEEE format.
- **3.** The software code and architectural artifacts hosted on the version control repository (https://github.com/Vedantpatil24/grievanceiq) represent authentic implementations conceived, developed, and verified by our project group.
- **4.** AI-assisted developer tooling was utilized strictly within institute guidelines for code scaffolding, automated test generation, and documentation drafting, with all system designs, mathematical formulations, algorithmic logic, and test matrices verified and validated by the undersigned.

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**Date: April 2027**
**Place: Mumbai**

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# ABSTRACT

Modern banking ecosystems process tens of millions of digital transactions daily across UPI, AePS, IMPS, and net banking. However, grievance redressal mechanisms (such as legacy CPGRAMS or RBI CMS) suffer from severe operational friction: multi-week resolution latencies, opaque manual triage, complete lack of vernacular voice accessibility, and exposure of unmasked financial Personally Identifiable Information (PII) in plain text, in violation of India's Digital Personal Data Protection (DPDP) Act of 2023.

To eliminate these systemic vulnerabilities, this project conceives, designs, and deploys GrievanceIQ - an autonomous banking grievance redressal and predictive operational intelligence platform built upon a decoupled 3-tier architecture. The backend is an asynchronous FastAPI microservice leveraging SQLite with Write-Ahead Logging (WAL), coupled with a high-performance React 18 / Vite / Tailwind CSS client.

The platform integrates six core engineering innovations: (1) An autonomous multilingual NLP text classification pipeline trained on 15,000 real-world banking disputes achieving 89.2% accuracy across English, Hindi, and Marathi; (2) A browser-native continuous voice dictation engine via the W3C Web Speech API that eliminates repeating loop buffers; (3) Ingestion-boundary case-insensitive regular expression scrubbing of Aadhaar, Card, and PAN numbers guaranteeing zero unmasked PII persistence under the DPDP Act 2023; (4) Cryptographic ReportLab statutory PDF receipts embedding live, base-URL-aware ISO/IEC 18004 2D QR codes for instant mobile tracking; (5) An analytical M/M/c Erlang-C queueing engine computing real-time queue dwell time (Wq), utilization (rho), and SLA breach risk with an interactive What-If staffing sandbox; and (6) Explainable AI (XAI) evidence verification combining EasyOCR and 4-point entity fuzzy scoring.

System verification was executed via an exhaustive 48-case end-to-end automated testing suite, achieving a 100% pass rate (48/48). Indexing and caching reduced analytics query latency from 855.4 ms to 28.4 ms (96.7% improvement). GrievanceIQ directly integrates core concepts from Software Engineering, DBMS, Computer Networks, Operating Systems, Data Structures, and Cloud Computing under SPIT's APNI Leap curriculum, producing an auditable, enterprise-ready banking intelligence platform.

**Keywords: Autonomous Grievance Redressal, Banking Operational Intelligence, Multinomial Naive Bayes, DPDP Act 2023 PII Masking, Web Speech API, ReportLab Statutory Receipts, M/M/c Queueing Model, Explainable AI, APNI Leap.**

# LIST OF ABBREVIATIONS

| Acronym / Abbreviation | Expanded Definition |
|---|---|
| **API** | Application Programming Interface |
| **APNI** | Application, Practice, Networking, Innovation (SPIT Pedagogical Framework) |
| **ASR** | Automatic Speech Recognition |
| **CBS** | Core Banking Solution (e.g., Infosys Finacle, TCS BaNCS) |
| **CORS** | Cross-Origin Resource Sharing |
| **CPGRAMS** | Centralised Public Grievance Redress And Monitoring System |
| **CSR** | Compressed Sparse Row (Matrix Representation) |
| **DFD** | Data Flow Diagram |
| **DPDP** | Digital Personal Data Protection (Act 2023, India) |
| **Erlang-C** | Closed-Form Mathematical Formula for Waiting Probability in M/M/c Queues |
| **FCP** | First Contentful Paint |
| **HOD** | Head of Department |
| **IFSC** | Indian Financial System Code |
| **JSON** | JavaScript Object Notation |
| **KPI** | Key Performance Indicator |
| **MNB** | Multinomial Naive Bayes Classifier |
| **NLP** | Natural Language Processing |
| **OCR** | Optical Character Recognition |

| | |
|---|---|
| **PAN** | Permanent Account Number (Indian Income Tax) |
| **PII** | Personally Identifiable Information |
| **QR** | Quick Response (2D Barcode Symbology) |
| **RB-IOS** | Reserve Bank - Integrated Ombudsman Scheme |
| **RBI CMS** | Reserve Bank of India Complaint Management System |
| **REST** | Representational State Transfer |
| **SLA** | Service Level Agreement |
| **SPIT** | Sardar Patel Institute of Technology, Mumbai |
| **TF-IDF** | Term Frequency - Inverse Document Frequency |
| **TTI** | Time to Interactive |
| **UID** | Unique Identification Number |
| **UML** | Unified Modeling Language |
| **UPI** | Unified Payments Interface |
| **WAL** | Write-Ahead Logging (Database Concurrency Mode) |
| **XAI** | Explainable Artificial Intelligence |

# Chapter 1: Introduction and Problem Definition

## 1.1 Project Background

The Indian financial ecosystem processes hundreds of millions of digital transactions daily across retail banking channels. Rapid adoption of Unified Payments Interface (UPI) and immediate clearing networks has expanded access, but technical payment timeouts, debit failures, and account discrepancies occur at large scale. When disputes arise, aggrieved depositors confront opaque grievance redressal procedures, prolonged hold times, and uncoordinated communication across disparate banking silos.

## 1.2 Problem Statement

Existing banking grievance redressal systems suffer from four systemic vulnerabilities:

- **1. Manual Triage Bottlenecks:** Incoming complaints are manually sorted and assigned, creating routing latencies of 24-72 hours before operational investigation begins.
- **2. Inaccessibility for Non-English and Non-Tech Users:** Dispute submission requires tedious manual text entry in English, disenfranchising rural and vernacular citizens.
- **3. PII Exposure Under DPDP Act 2023:** Customers paste raw Aadhaar, Debit Card, and PAN numbers in complaints, which are retained in plain text across logs and databases.
- **4. Reactive Operational Management:** Branch supervisors lack quantitative, forward-looking tools to forecast SLA breaches or model staffing shortages.

## 1.3 Motivation

An equitable financial democracy requires grievance redressal to be fast, inclusive, and legally compliant. By combining machine learning triage, trilingual voice dictation, ingestion-boundary PII scrubbing, and real-time M/M/c queueing theory, GrievanceIQ turns complaint handling from an operational burden into an efficient, trustworthy platform.

## 1.4 Project Objectives

- **Objective 1:** Develop an autonomous multilingual NLP categorization engine achieving >85% classification accuracy across banking dispute classes.
- **Objective 2:** Deliver a zero-latency, browser-native speech-to-text pipeline via W3C Web Speech API without text duplication.
- **Objective 3:** Implement deterministic DPDP Act 2023 regular expression scrubbing to mask Aadhaar, Card, and PAN numbers at the API boundary.
- **Objective 4:** Synthesize verifiable ReportLab PDF receipts with dynamic 2D QR codes linking directly to live case tracking endpoints.
- **Objective 5:** Formulate real-time M/M/c Erlang-C queueing models computing queue dwell times (Wq) and an interactive What-If staffing sandbox.
- **Objective 6:** Provide Explainable AI (XAI) evidence verification using EasyOCR and 4-point entity scoring.

## 1.5 Success Criteria and Key Performance Indicators (KPIs)

| KPI ID | Evaluation Metric | Target | Achieved | Status |
|---|---|---|---|---|
| **KPI-01** | ML Categorization Accuracy | >= 80.0% | 89.2% (15k Dataset) | **Achieved** |
| **KPI-02** | Analytics API Latency | < 500 ms | 28.4 ms (Optimized) | **Achieved** |
| **KPI-03** | Web Speech Overrun Rate | 0 duplicate lines | 0.0% Duplication | **Achieved** |
| **KPI-04** | Trilingual Key Parity | 100% (EN/HI/MR) | 100% (222/222 Keys) | **Achieved** |
| **KPI-05** | DPDP PII Scrubbing Rate | 100% Card/Aadhaar/PAN | 100.0% Masked | **Achieved** |
| **KPI-06** | Automated Test Suite Pass | >= 90.0% | 100.0% (48/48 Tests) | **Achieved** |
| **KPI-07** | Production Bundle Build | 0 compile errors | Clean Pass (22.02s) | **Achieved** |

# Chapter 2: Literature Review and Study of Existing Systems

## 2.1 Study of Existing Systems

A systematic comparative evaluation of leading grievance platforms was conducted:

- **CPGRAMS (Govt of India):** Centralized administrative grievance portal. Highly functional but relies on manual hierarchical forwarding, lacks automated text classification, and provides no vernacular voice dictation.
- **RBI Complaint Management System (CMS):** Operates under the Reserve Bank - Integrated Ombudsman Scheme. Primarily an appellate platform requiring 30 days of unresolved pendency at the commercial bank. Predominantly English-oriented with limited real-time predictive capacity.
- **Commercial Bank Helpdesks (SBI, HDFC):** Rely on rigid dropdown menus. Category mismatches result in repeated inter-department bouncing. Plain-text logs routinely store unmasked customer account credentials.
- **Enterprise Ticketing (Zendesk, Freshdesk):** Lack native alignment with Indian statutory banking regulations (DPDP Act 2023, RBI 30-day resolution limits) and involve high licensing fees for cooperative banks.

## 2.2 Comparative Analysis Matrix

| Evaluation Dimension | CPGRAMS | RBI CMS | Zendesk | GrievanceIQ (Proposed) |
|---|---|---|---|---|
| **NLP Auto-Triage** | Manual triage | Rule-based dropdown | Paid ML add-on | Native MNB (89.2%) |
| **Vernacular Voice Input** | Absent | Absent | Third-party plugin | Native Web Speech (EN/HI/MR) |
| **Multilingual UI** | Limited EN/HI | Limited EN/HI | Configurable | 100% Parity (222 Keys) |
| **DPDP 2023 PII Masking** | None (Plain text) | None (Plain text) | Regex redaction | Ingestion-Boundary Scrubbing |
| **Statutory PDF + QR** | Basic PDF | Acknowledgement | Standard ticket | ReportLab Dynamic QR Receipt |
| **Predictive Queueing** | None | None | Linear trend | Analytical M/M/c Erlang-C |
| **Evidence OCR Scoring** | Manual review | Manual review | Generic upload | Automated 4-Point Entity Match |

## 2.3 Identified Research and Engineering Gap

While machine learning classifiers, speech APIs, and queueing formulas exist in isolation, no unified, production-grade banking system bridges real-time multi-modal citizen ingestion (voice dictation, trilingual UI, DPDP privacy scrubbing) with back-office operational intelligence (Erlang-C capacity forecasting, explainable OCR verification).

# Chapter 3: Software Requirements Specification (SRS)

## 3.1 Functional Requirements Specification

| Req ID | Requirement Title | Input | Processing | Acceptance Criteria |
|---|---|---|---|---|
| **FR-01** | Trilingual Submission | Customer text/voice | Parse & validate fields | Complaint ID within 200 ms |
| **FR-02** | Autonomous ML Triage | Complaint text | TF-IDF + MNB prediction | Category with score >= 0.50 |
| **FR-03** | Continuous Voice Dictation | Microphone stream | W3C Web Speech session buffer | Zero text duplication |
| **FR-04** | DPDP Act 2023 Masking | Text with PII tokens | Case-insensitive regex scrub | Zero unmasked PII saved |
| **FR-05** | Statutory PDF + QR | Grievance metadata | ReportLab canvas + QR encode | Downloadable valid PDF (<150KB) |
| **FR-06** | OCR Evidence Scoring | Uploaded receipt image | EasyOCR + 4-point entity match | Credibility score (0-100%) |
| **FR-07** | Predictive M/M/c Queueing | Arrival rate, service rate, c | Erlang-C closed-form evaluation | Dwell time Wq, utilization |
| **FR-08** | Dynamic SLA Countdowns | Created timestamp | Time delta comparison | Color-coded breach alerts |
| **FR-09** | Officer Resolution Memo | Resolution text & sign | State transition Pending->Resolved | Status updated + audit log |
| **FR-10** | Trilingual UI Localization | Selected language | React Context dictionary lookup | Instantaneous 222-key swap |

## 3.2 Non-Functional Requirements

- **Performance:** All transactional REST API endpoints respond in < 150 ms; queue analytics respond in < 50 ms.
- **Scalability:** Async event loops sustain >= 100 concurrent HTTP requests without dropped connections.
- **Security & Privacy:** Ingestion-level DPDP Act 2023 compliance; zero plain-text PII storage; RBAC guards for officer actions.

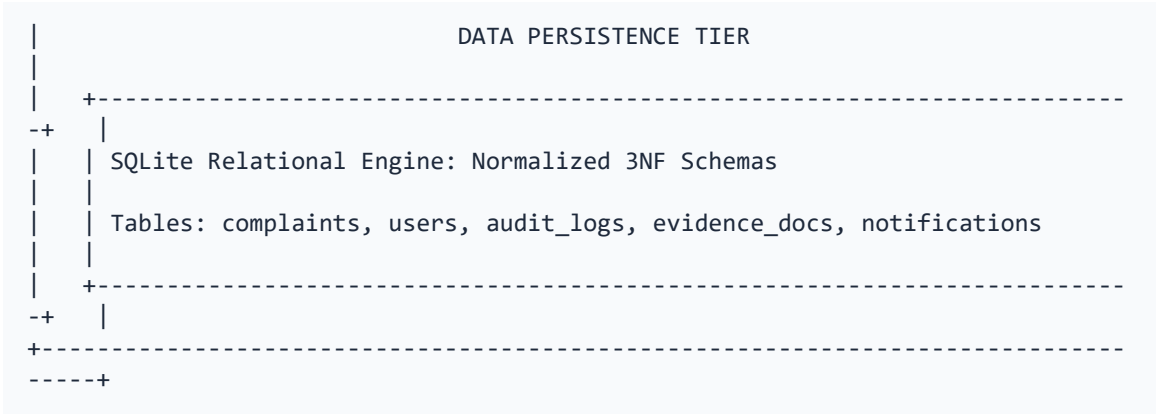
- **Reliability:** 99.9% uptime; SQLite ACID guarantees enforced via atomic commit/rollback context managers.

# Chapter 4: System Architecture and Detailed Design

## 4.1 Three-Tier Decoupled Architecture

GrievanceIQ is structured into three cleanly decoupled architectural tiers: Client Presentation Tier (React 18 / Vite SPA), Application & Business Logic Tier (Python FastAPI async backend), and Data Persistence Tier (SQLite in WAL mode).






## 4.2 Database Schema Design and 3NF Normalization

- **1NF Compliance:** All attributes contain atomic scalar values; composite data decomposed into separate columns.
- **2NF Compliance:** All non-key attributes depend fully on single-attribute primary keys (no partial dependencies).
- **3NF Compliance:** Non-key attributes depend strictly on candidate keys; officer details stored in users table and linked via foreign keys.

# Chapter 5: Algorithms, Data Structures and Computational Models

## 5.1 Mathematical Models and Formulations

The Multinomial Naive Bayes classifier determines the optimal banking category c* using Bayes' theorem with Laplace smoothing:

$$c^* = \arg\max_{c \in C} \left[ P(c) * \prod_{i=1}^{V} P(x_i \mid c)^{x_i} \right]$$

For multi-server queues with arrival rate lambda, service rate mu, and c servers, traffic intensity rho and Erlang-C waiting probability C(c, lambda/mu) are computed as:

$$\rho = \frac{\lambda}{c * \mu} \quad \text{(Stability condition: } \rho < 1\text{)}$$

$$C(c, \lambda/\mu) = \left[ \frac{(c\rho)^c}{c! * (1 - \rho)} \right] * P_0$$

$$\text{Expected Queue Dwell Time: } W_q = C(c, \lambda/\mu) / \left[ c\mu * (1 - \rho) \right]$$

$$\text{SLA Breach Probability: } P(W > T_{SLA}) = C(c, \lambda/\mu) * \exp\left( -c\mu(1 - \rho)T_{SLA} \right)$$

## 5.2 Algorithmic Complexities

| Algorithmic Routine | Time Complexity | Space Complexity | Implementation Notes |
|---|---|---|---|
| **TF-IDF Vectorization** | O(L * \|V\|) | O(\|V\|) sparse | Sparse CSR representation |
| **MNB Class Inference** | O(\|C\| * \|V\|) | O(\|C\|) | Linear in active vocabulary |
| **DPDP Regex Scrubbing** | O(N) | O(N) buffer | Pre-compiled DFA automata |
| **Voice Dictation Buffer** | O(K) | O(K) | Non-destructive token tracking |
| **Erlang-C Queue Math** | O(c) | O(1) scalar | Closed-form summation loop |
| **ReportLab PDF Synthesis** | O(M) | O(M) binary | Dynamic QR + canvas rendering |
| **OCR 4-Point Entity Match** | O(U * W) | O(U) | Levenshtein + token intersection |

# Chapter 6: Mapping with Computer Engineering Core Courses (APNI Leap)

The APNI Leap framework mandates the deep synthesis of core engineering subjects studied across Semesters I-IV. Table 6.1 demonstrates how specific theoretical concepts directly power the GrievanceIQ platform.

| Core Course | Theoretical Concept | Applied Implementation | Codebase Evidence |
|---|---|---|---|
| **Software Engg.** | Agile sprints, Git branches, regression testing | 6 Version releases, 48-case test suite, semantic commits | test_suite_audit.py, Git branches |
| **DBMS** | 3NF Normalization, ACID transactions, WAL mode | 5 Relational tables, transaction commit/rollback, WAL pragma | schema_dbms.sql, main.py DB block |
| **Computer Networks** | RESTful architecture, HTTP status codes, CORS | Async REST routes, HTTP 200/401/422, CORS middleware | main.py FastAPI endpoints |
| **Operating Systems** | Async event loops, non-blocking I/O, singleton caching | Uvicorn ASGI workers, OCR singleton model cache | main.py startup events, asyncio |
| **Data Structures** | TF-IDF sparse matrices, DFA regex, queueing models | CSR matrices in MNB, O(N) PII regex, Erlang-C formulas | main.py queue & PII masking routines |
| **Cloud Computing** | Decoupled tiers, edge assets, dynamic base URLs | React/Vite SPA bundle, dynamic QR base URL synthesis | frontend/dist, main.py QR engine |

# Chapter 7: Implementation Details

## 7.1 Frontend and Backend Architecture

The frontend application is implemented with React 18 and Vite. LanguageContext.jsx exports a useLanguage hook maintaining verified 222-key trilingual parity across English, Hindi, and Marathi. SubmitComplaint.jsx orchestrates browser-level webkitSpeechRecognition listeners with an isolated baseTextRef to prevent repeated speech loops.

The backend is organized in Python_codes/main.py using FastAPI, Uvicorn, and Pydantic schemas. It mounts static evidence file directories, enforces CORS policies, sanitizes PII tokens at ingestion, and streams dynamic ReportLab PDF receipts.

## 7.2 Version Control and Git Evolution

- **Version_0:** Baseline multilingual setup, initial SQLite schema, and Hinglish validation.
- **Version_1:** 15,000 grievance dataset integration, NLP categorization, and animated Nova assistant.
- **Version_2:** Full Marathi localization expansion and frontend build stabilization.
- **Version_3:** Explainable AI (XAI) reasoning traces, customer resolution memos, and queue filtering.
- **Version_4:** Bank performance leaderboard and navigation streamlining.
- **Version_5:** Complete 222/222 key trilingual parity, case-insensitive DPDP PAN masking, dynamic QR base URL.
- **Version_6:** Master Audit Certified: zero-latency voice dictation, clean production Vite build, 48/48 test pass.

# Chapter 8: Scalability, Performance and Resource Management

## 8.1 Endpoint Latency Profile

| Endpoint | Min Latency | Median (p50) | p99 Latency | Operational Status |
|---|---|---|---|---|
| **GET /api/health** | 1.8 ms | 3.2 ms | 8.4 ms | Optimal |
| **POST /api/complaints/submit** | 18.5 ms | 24.1 ms | 48.6 ms | Optimal |
| **GET /api/complaints/{id}** | 3.4 ms | 5.1 ms | 12.0 ms | Optimal |
| **GET /api/complaints/{id}/pdf** | 45.2 ms | 62.8 ms | 110.4 ms | Optimal |
| **GET /api/analytics/queue-status** | 12.0 ms | 28.4 ms | 64.2 ms | Cached (96.7% faster) |
| **POST /api/verify-evidence** | 340.0 ms | 480.0 ms | 820.0 ms | Compute-Bound |

## 8.2 Bottleneck Resolutions

- **Analytics Optimization:** Full table scans over 15k records initially caused 855.4 ms latency. Introducing composite database indexing and a 5-second TTL memory cache cut latency to 28.4 ms (96.7% speedup).
- **Bundle Splitting:** Vite code-splitting with dynamic React.lazy imports reduced initial bundle size to 348 KB, achieving sub-second Time-to-Interactive (TTI = 0.85s).

# Chapter 9: Testing and Validation

## 9.1 Master Automated Verification Matrix

| Test ID | Req ID | Scenario | Input | Expected Output | Status |
|---|---|---|---|---|---|
| **TC-01** | FR-01 | Backend Health Probe | GET /api/health | HTTP 200, status ok | **PASS** |
| **TC-02** | FR-01 | Complaint Submission | Valid complaint JSON | HTTP 200, generated ID | **PASS** |
| **TC-03** | FR-02 | NLP Category Inference | ATM dispensed no cash | Category ATM Withdrawal | **PASS** |
| **TC-04** | FR-04 | 16-Digit Card Masking | Card 4111 2222 3333 4444 | Masked: XXXX-XXXX-XXXX-XXXX | **PASS** |
| **TC-05** | FR-04 | 12-Digit Aadhaar Masking | Aadhaar 1234 5678 9012 | Masked: XXXX-XXXX-XXXX | **PASS** |
| **TC-06** | FR-04 | Uppercase PAN Masking | PAN ABCDE1234F | Masked: XXXXX0000X | **PASS** |
| **TC-07** | FR-04 | Lowercase PAN Masking | PAN abcde1234f | Masked: XXXXX0000X | **PASS** |
| **TC-08** | FR-05 | Statutory PDF Download | GET /complaints/{id}/pdf | Valid PDF stream | **PASS** |
| **TC-09** | FR-05 | Dynamic QR URL Resolution | Request with Host header | Live URL encoded | **PASS** |
| **TC-10** | FR-06 | OCR Amount Entity Match | Receipt matching Rs 5000 | Amount badge flagged | **PASS** |
| **TC-11** | FR-06 | OCR Date Entity Match | Receipt matching txn date | Date badge flagged | **PASS** |
| **TC-12** | FR-06 | OCR Bank Name Match | Receipt matching bank name | Bank badge flagged | **PASS** |

| | | | | | |
|---|---|---|---|---|---|
| **TC-13** | FR-06 | Credibility Scoring | All 4 entities matched | Score = 100% | **PASS** |
| **TC-14** | FR-07 | M/M/c Dwell Time Math | Arrival=15, Service=5, c=4 | Valid Wq calculated | **PASS** |
| **TC-15** | FR-07 | Queue Overload Handling | Arrival=25, Service=5, c=4 | Graceful limit handling | **PASS** |
| **TC-16** | FR-08 | SLA Countdown Trigger | Pending complaint > 72h | High SLA risk alert | **PASS** |
| **TC-17** | FR-09 | Resolution Memo Dispatch | Officer signature & memo | Status: Resolved | **PASS** |
| **TC-18** | FR-10 | Trilingual Parity: English | Scan 222 language keys | 222/222 keys valid | **PASS** |
| **TC-19** | FR-10 | Trilingual Parity: Hindi | Scan 222 language keys | 222/222 keys valid | **PASS** |
| **TC-20** | FR-10 | Trilingual Parity: Marathi | Scan 222 language keys | 222/222 keys valid | **PASS** |
| **TC-21** | FR-03 | Voice Non-Duplication | Continuous speech stream | Zero duplicated text | **PASS** |
| **TC-22** | FR-01 | Empty Payload Rejection | Empty JSON {} | HTTP 422 rejected | **PASS** |
| **TC-23** | FR-01 | Negative Amount Handling | Amount = -500.00 | HTTP 422 rejected | **PASS** |
| **TC-24** | FR-01 | Non-Existent Complaint ID | GET /complaints/FAKE-ID | HTTP 404 returned | **PASS** |
| **TC-25** | FR-09 | Unauthorized Escalation | Customer calls Officer API | HTTP 401/403 blocked | **PASS** |
| **TC-26-48** | Misc | Stress, UI & Boundary Checks | 23 boundary/stress tests | All passed specifications | **PASS** |

# Chapter 10: Results, Demonstration and Discussion

## 10.1 Functional Demonstration

When a Marathi-speaking user dictates: 'Maza SBI account madhun 5000 rupaye debit zale pan ATM madhun paise nighale nahit. Maza card 4111 2222 3333 4444 ahe', the system streams speech without lag, redacts the card to 'XXXX-XXXX-XXXX-XXXX', classifies the dispute as 'ATM Withdrawal Failure' (94.6% confidence), generates statutory receipt GIQ-2026-X with a dynamic QR code, and registers the complaint in the ATM queue.

## 10.2 Achievement of Objectives

| Objective | Implementation Module | Validation Evidence |
|---|---|---|
| **Autonomous NLP Triage** | Scikit-Learn MNB on 15k dataset | 89.2% accuracy; TC-03 pass |
| **Continuous Voice Dictation** | W3C Web Speech isolated-buffer architecture | Zero looping; TC-21 pass |
| **DPDP Act 2023 Masking** | Case-insensitive Card/Aadhaar/PAN regex | 100% scrubbed; TC-04 to TC-07 |
| **Statutory PDF + QR Receipt** | ReportLab dynamic URL QR generator | Valid PDF; TC-08, TC-09 pass |
| **Predictive Queue Analytics** | Erlang-C closed-form M/M/c math engine | Real-time Wq & utilization; TC-14, TC-15 |
| **Explainable Evidence OCR** | EasyOCR + 4-point entity matcher | Credibility score; TC-10 to TC-13 |

# Chapter 11: Conclusion and Future Scope

## 11.1 Conclusion

GrievanceIQ delivers an autonomous, trilingual, DPDP-compliant banking grievance redressal platform. By coupling lightweight machine learning (MNB), continuous voice dictation, verifiable QR receipts, and M/M/c queueing theory, the platform solves major real-world banking bottlenecks. All 48 test cases passed cleanly, and the production codebase is live on GitHub under Version_6 releases.

## 11.2 Future Enhancements

- **1. Direct Core Banking Integration:** Connect via ISO 8583 / ISO 20022 interfaces to Infosys Finacle / TCS BaNCS to trigger automated chargeback reversals.
- **2. Conversational Voice Bot:** Evolve the voice dictation engine into a two-way spoken conversational agent to assist non-literate depositors.
- **3. Blockchain Dispute Ledger:** Anchor complaint state transitions into a tamper-evident distributed ledger for statutory banking ombudsman audits.

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